Table 1 Satiety hormones (mean ± standard error of the mean) in plasma of fasting dogs fed the low, medium and high RS diets.

Satiety hormone	Low RS	Medium RS	High RS	*P* (F)
Ghrelin, ln	1.565 ± 0.0229	1.597 ± 0.0226	1.582 ± 0.0227	0.1295
Leptin, ln	5.28 ± 0.2299	5.51 ± 0.2290	5.38 ± 0.2299	0.1263
GIP, ln	1.068 ± 0.1176	1.035 ± 0.1176	1.052 ± 0.1168	0.9221
GLP-1, ln	0.974 ± 0.1556	1.144 ± 0.1343	1.046 ± 0.1292	0.3291
Glucagon, ln	1.330 ± 0.0284	1.337 ± 0.0283	1.336 ± 0.0283	0.7221
Insulin, ln	4.07^{b} ± 0.1616	4.31^{ab} ± 0.1599	4.46^{a} ± 0.1599	0.0237
PP, ln	3.97 ± 0.1522	3.96 ± 0.1507	4.06 ± 0.1507	0.6649
PYY, ln	4.63 ± 0.0555	4.66 ± 0.0551	4.67 ± 0.0551	0.6029

Satiety hormones measured as pg/mL and converted to natural log.
Tukey correction on SAS.

Keywords: dog, resistant starch, satiety

PRODUCTION, MANAGEMENT AND ENVIRONMENT

PSXI-4 Beauveria bassiana to control the mealworm Alphitobius diaperinus in chicken barns.

Martin Nadeau[1], Silvia Todorova[1], Carl Julien[2], [1]*Anatis Bioprotection,* [2]*CRSAD*

One of the most common insects in poultry houses is the mealworm Alphitobius diaperinus, also known as the darkling beetle. Immature (larvae) and adults feed on seeds spilled on the ground or from poultry droppings. In addition to structural damage, this insect can also be a vector of diseases that can affect birds. In addition, the indigestible shell of the insect can cause digestion problems for birds, and thus affect the profitability of production. It is considered impossible to eliminate the populations of mealworms, and in organic poultry production, the control options are limited. The mycoinsecticide BioCeres WP, based on the fungus Beauveria bassiana, and which controls aphids, has been proposed to control the mealworm A. diaperinus in chicken barns. 450 adult mealworms from a commercial chicken barn were randomly distributed in 9 containers (1 m^2) containing 10 cm of wood shavings litter and chicken feed. BioCeres WP was applied by spraying two applications of 40 ml of 10 g/L solution (representing 4 x 10^9 conidia per m^2 per application). Mealworms were housed during a broiler batch at the chicken barn they were harvested from and received one of the following treatments: 1) control, water spraying; 2) BioCeres WP treatment. 3 repetitions for control and 6 for BioCeres WP were performed. BioCeres WP increased the mortality rate of mealworms (+101%, $P < 0.02$; 17.8% vs. 8.6%), compared to untreated mealworms, 28 days after treatment. Mortality rates were affected from day 14 until day 28 after treatment. A trial using 640 broilers in 32 pens revealed no adverse effect on growth performance or health. In summary, BioCeres WP could be involved in a biological strategy to control the mealworm Alphitobius diaperinus in chicken barns.

Keywords: darkling beetle, pest, broilers